

A Hybrid Customer Segmentation and Repurchase Prediction Framework Using RFMT Features and Machine Learning

Phuong-Vi Tran Nguyen^{1*}, Ngoc-Giao Nguyen¹, Hong-Ngoc Nguyen Thi¹

¹Department of Business Information Technology, University of Economics Ho Chi Minh City, Ho Chi Minh City, Vietnam.

*Corresponding Author: (Email: phuongvi138bp@gmail.com)

Abstract. This paper proposes a hybrid customer segmentation and repurchase prediction framework that integrates RFMT-based clustering with machine learning classification for e-commerce customer analytics. Using the online retail dataset containing over one million transaction records, the study applies multiple clustering algorithms, including K-Means, Fuzzy C-Means, Spectral Clustering, and Mean Shift, to identify distinct customer segments. The research further evaluates whether building separate predictive models for each customer segment can outperform a traditional global prediction approach. To address the class imbalance problem in customer repurchase prediction, the SMOTE-Tomek technique was incorporated during the training process. Experimental results show that cluster-based prediction models consistently outperform the global baseline, particularly after data balancing. Among all evaluated approaches, the hybrid K-Means-XGBoost model combined with SMOTE-Tomek balancing method achieved the best overall performance, with an F1-score of 0.5901 and a ROC-AUC of 0.8247. Clustering results reveal different behaviour patterns among customer groups, highlighting the strategic importance of high-value segments such as VIPs and Potential Loyalists. The study demonstrates that integrating customer segmentation with predictive modeling improves both model interpretability and business decision-making for targeted retention strategies in the retail sector.

Keywords: Customer segmentation; RFMT analysis; Repurchase prediction; Hybrid model; E-commerce analytics

1 Introduction

In recent years, the internet explosion has encouraged the rapid growth of the e-commerce industry, resulting in an increasingly competitive market (Ahmed et al., 2024). In this landscape, consumers have tended to easily switch between service platforms due to the abundance of alternative options (Martínez et al., 2020). To optimize financial efficiency, businesses now focus their investments on existing customers, as the return on investment (ROI) from retention strategies consistently outperforms the cost of new customer acquisition (Jahromi et al., 2014). As a result, accurately forecasting customer repurchase behavior is equally vital, as it allows businesses to pinpoint potential repeat buyers and implement targeted strategies (Verma et al., 2025). Despite the growing attention to prediction systems, many traditional methods still have limitations that need to be addressed. First, customer

heterogeneity is overlooked. Previous studies applied a single prediction model to an entire dataset, which fails to capture the behavioral patterns in different customer segments (Tang et al., 2020). Many previous studies still handle customer segmentation and churn or repurchase prediction separately, instead of integrating them into a specific framework to optimize overall performance (Ahmed et al., 2024). The classic static RFM (Recency, Frequency, Monetary) model fails to reflect the temporal dynamics and shopping rhythm of consumers (Zhou et al., 2021). The persistent challenge of data imbalance is often not fully addressed, leading to biased models (Chawla et al., 2002). To address these gaps, this study proposes a hybrid cluster-then-predict framework. The research aims to answer a key question: “Does building separate repurchase prediction models for each segment yield better performance than using a single model for the entire dataset?” By integrating unsupervised clustering algorithms with classification algorithms and applying the SMOTE-Tomek technique for data balancing, this approach aims to improve the model performance and provide a more detailed identification of at-risk customers. The main finding of this study is demonstrating that segment-based modeling delivers superior performance compared to overall models by identifying complex patterns. Simultaneously, it shows the effectiveness of the SMOTE-Tomek technique in mitigating bias and improving evaluation metrics. Finally, establishing a comprehensive integrated system helps the retail and e-commerce businesses to optimize resources and implement effective customer retention strategies.

2 Literature Review

2.1 RFM and RFMT Analysis in the Retail Sector

RFM model (Recency, Frequency, Monetary), introduced by Hughes (2005), is a robust framework for facilitating behaviour analysis and customer segmentation within the retail sector. This model evaluates customer value based on three core dimensions:

- Recency (R): the time elapsed since the last transaction
- Frequency (F): the total number of purchases within a specific period
- Monetary (M): the total monetary value spent by the customer

Rahul et al. (2021) indicated that integrating R, F, and M metrics with clustering algorithms significantly increased the accuracy of customer segmentation compared to other traditional methods. Numerous scholars have extended the traditional RFM model by incorporating additional variables to improve its predictive power. For example, Wu et al. (2022) considered C (Add to cart) and V (Add favorites) as essential factors into an RFMCV model, which was coupled with K-medoids to yield better clustering results. Zhou et al. (2021) proposed an extended RFMT with a temporal dimension (T), representing the inter-purchase time. This T measures the average time between transactions, providing deep insights into purchasing cycles and customer loyalty patterns. Building upon these findings, Ullah et al. (2023) applied the RFMT model to a large-scale e-commerce dataset in Pakistan. These studies clearly demonstrate that the performance of RFMT is superior to that of traditional RFM in identifying clusters with distinct characteristics. Thus, this factor becomes a significant contribution to marketing strategies as well as customer relationship management. Therefore, this research adopts a hybrid approach combining RFMT analysis with multiple clustering algorithms, aiming to enhance segmentation efficiency and provide actionable managerial insights.

2.2 Customer Clustering Algorithms

Rahul et al. (2021) characterize K-means as a fundamental yet robust algorithm that partitions data by minimizing the Euclidean distance between data points and their centroids. Gankidi et al. (2022) also mentioned that this algorithm is favored because of its simplicity and scalability across diverse dataset sizes. However, a significant limitation of traditional K-means is its sensitivity to outliers, which can undermine the stability of clustering outcomes (Wu et al., 2022). To mitigate arbitrary or subjective bias in selecting the number of clusters (k), researchers implement various validation metrics. For instance, Nguyen and Bui (2024) and Ullah et al. (2023) utilize the Elbow method and Silhouette score to identify the most stable cluster configuration. Similarly, Ding et al. (2022) rely on the Davies-Bouldin index and Calinski-Harabaz index to determine the optimal number of clusters, ensuring that the segments are well-separated and dense. In contrast to the hard partitioning logic of K-means, Fuzzy C-Means introduces soft clustering, where each data point is assigned a membership degree across multiple clusters. Ding et al. (2022) proposed integrating clustering into hybrid models to address data ambiguity and uncertainty before the predictive stage. For instance, in the telecommunications sector, Praseeda and Shivakumar (2021) leveraged FCM-derived variables to enhance customer retention strategies, demonstrating the algorithm's utility in capturing complex behavioral overlaps. Spectral clustering excels in capturing intricate structures that K-means often fails to identify (Wu et al., 2022). In their study on standard datasets, Wu et al. (2022) found that while the algorithm provides high-quality results, its processing time is generally longer than that of other clustering methods. Mean Shift is highly regarded in data mining for its density-based mode-seeking capability, allowing it to automatically deduce the underlying cluster structure without relying on a pre-defined number of clusters (Chakraborty et al., 2021). Instead, it automatically identifies high-density areas to form groups. This makes the algorithm very effective for businesses dealing with customers who exhibit complex behaviors, significant data noise, or arbitrary segment shapes that are hard to define at the start.

2.3 Repurchase Prediction

In today's competitive business markets, the cost of acquiring a new customer is estimated to be higher than the cost of retaining an existing one (Martínez et al., 2020). Consequently, many businesses are investing heavily in analysing customer purchase intentions and repurchase behavior to design more effective retention strategies (Ekanayake and De Alwis, 2025). Logistic Regression is a widely utilized classification algorithm, valued for its simplicity and probabilistic interpretability (Almuqren et al., 2021). Li and Li (2019) integrated this algorithm with XGBoost in a customer retention context, achieving a prediction accuracy of approximately 85%. However, research indicates that while Logistic Regression is easy to implement, it often yields lower recall metrics compared to other algorithms, particularly in imbalanced customer behavior prediction tasks (Ekanayake and De Alwis, 2025). Random Forest is highly regarded for its effectiveness in handling large datasets, managing noise, and mitigating overfitting (Gan, 2022). In their study, Nguyen and Bui (2024) demonstrated that Random Forest outperformed XGBoost and k-NN in identifying customers with complex behavioural patterns. Supporting this evidence, Ekanayake and De Alwis (2025) applied Random Forest to purchase and retention prediction, reporting

significant improvements in classification performance. In the e-commerce sector, Gradient Boosting algorithms consistently show superior performance in terms of prediction accuracy and computational efficiency (Gan, 2022). Specifically, Gan (2022) observed that XGBoost significantly reduced the misclassification between loyal customers and potential churners. Dua et al. (2025) showed that the XGBoost algorithm achieves near-perfect accuracy when utilized to classify customers following the clustering phase. Additionally, Liu et al. (2022) presented a hybrid model combining K-medoids and Gradient Tree Boosting, which achieved 96% accuracy, outperforming single-classifier approaches.

2.4 Hybrid Clustering-Prediction Model

Recent studies suggest that instead of applying a global model to a dataset with low homogeneity, performing customer segmentation prior to forecasting is more effective (Dua et al., 2025). Common implementation strategies include integrating cluster labels as additional features (Bose and Chen, 2009) or developing parallel predictive models for each identified segment (Tang et al., 2020). According to Verma et al. (2025), using the Cluster ID as a predictive feature yields superior accuracy in repurchase forecasting. This identifier emerges as the most significant predictor, surpassing traditional frequency and monetary metrics, by effectively embedding group-specific behavioral contexts that global models fail to capture. Xiahou and Harada (2022) applied a cluster-then-predict approach and observed significant improvements in predictive accuracy. Ahmed et al. (2024) further support that this methodology effectively captures nuanced shopping patterns in the e-commerce sector. Building on these findings, this research adopts a hybrid model combining unsupervised and supervised algorithms to enhance prediction performance compared to traditional single-stage models.

2.5 Handling Data Imbalance With SMOTE-Tomek

Data imbalance is a widely recognized challenge in customer behavior prediction tasks, including churn and repurchase prediction, where the minority class is significantly smaller than the majority (Chawla et al., 2002). To address this, SMOTE-Tomek is applied to generate synthetic samples for the minority class while eliminating noisy data points that typically reside on the class boundaries. The SMOTE technique increases the minority class size by interpolating between existing data points and their nearest neighbors within the feature space (Chawla et al., 2002). Simultaneously, Tomek Links detects and removes noisy samples on the boundary, specifically majority class points that are too close to the minority class, to reduce class overlap (Tomek, 1976). By integrating these two techniques, SMOTE first balances the class distribution, followed by Tomek Links to refine the decision boundary by removing noise. Recent research by Gan (2022) confirms that SMOTE-Tomek significantly improves the predictive performance of XGBoost or Random Forest, especially when applied to complex e-commerce or banking datasets.

3 Research Methodology

3.1 Overview Of The Research Framework

This study establishes an empirical analysis framework consisting of five stages: (1) Data collection, (2) Preprocessing, (3) Feature engineering, (4) Model building, and (5) Evaluation.

As shown in Fig. 1, after the RFMT feature engineering phase, the workflow splits into parallel tracks. Track A constructs a global classification baseline under imbalanced and balanced conditions. Conversely, track B applies a hybrid approach where customer segmentation is set up prior to per-segment classification. This workflow aims to optimize performance by personalizing prediction models for each customer segment.

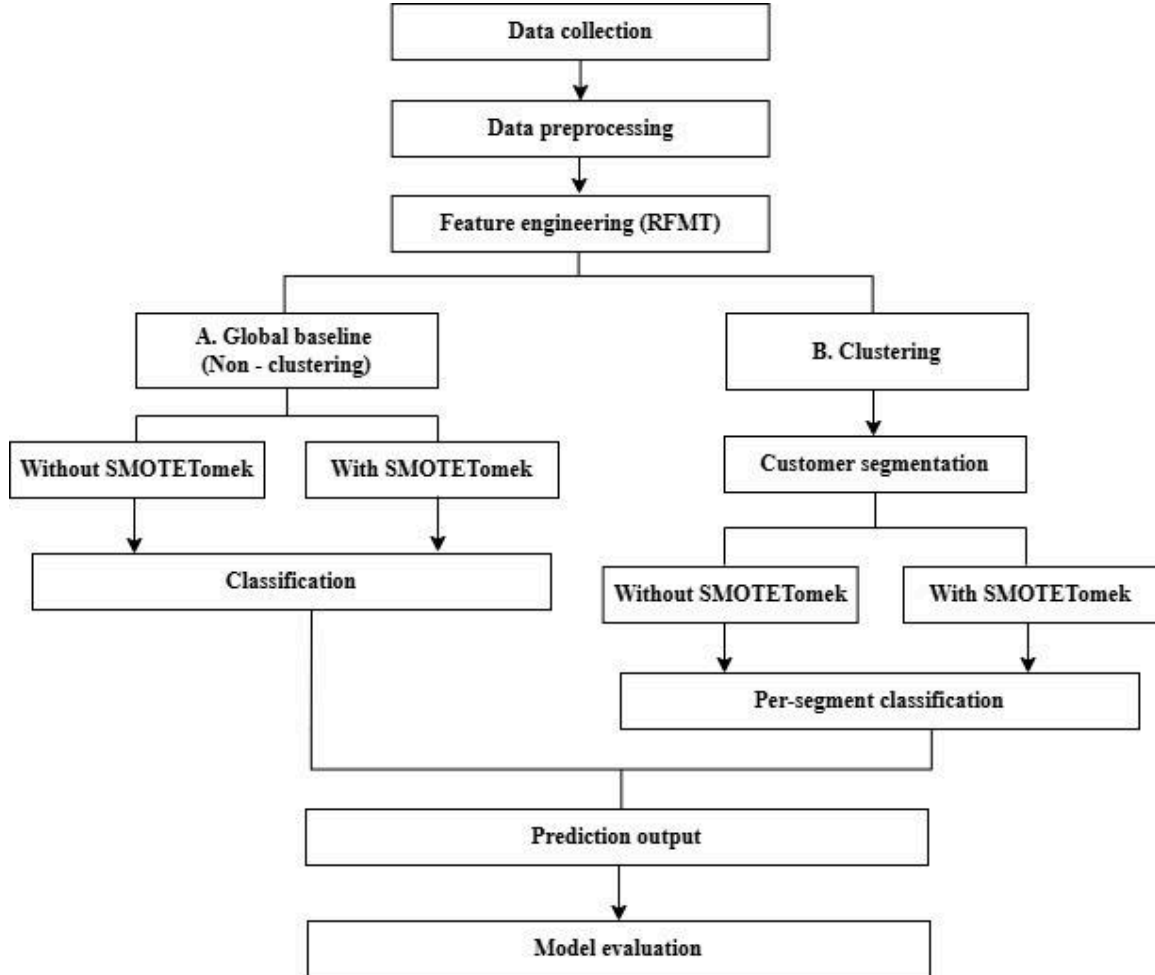


Fig. 1. Methodological workflow of the study

3.2 Data Collection And Description

The dataset used in this study is the Online Retail dataset published by Chen (2015), which was collected from a UK-based online retail company between December 2009 and December 2011. The dataset was obtained from the UCI Machine Learning Repository, a widely recognized academic data source commonly used in research on data mining and customer behavior analysis. To provide a clear overview, Table 1 details the variables extracted from the raw data, along with their types and specific roles in the process. For instance, identification keys like CustomerID and Invoice are essential for tracking transactions, while quantitative attributes such as Quantity, Price, and InvoiceDate are used as the foundation for calculating the behavioral RFMT features.

Table 1. Dataset description

No.	Variable Name	Data Type	Meaning
1	Invoice	Nominal	Transaction identifier. Invoices starting with 'C' indicate cancellations.
2	StockCode	Nominal	Item code. Used to identify product diversity.
3	Description	Nominal	Product name. Used primarily during data cleaning to exclude non-commercial records.
4	Quantity	Numeric	Product quantity per invoice. Negative values represent cancellations. A core component for Monetary (M) calculation.
5	InvoiceDate	Datetime	The date and time of a transaction. Important for engineering for Recency (R) and Inter-purchase Time (T).
6	Price	Numeric	Unit price (in GBP). Combined with Quantity to determine the total order value.
7	Customer ID	Nominal	A unique customer identifier. Primary key to aggregate raw data into RFMT features.
8	Country	Nominal	The country of customer residence.

3.3 Data Preprocessing And Feature Engineering

3.3.1 Data Cleaning And Initial Outlier Treatment

To eliminate noise and ensure cross-model consistency, raw records containing missing CustomerID, duplicate transactions, non-retail stock codes, and canceled transactions, including negative Quantity and zero Price were removed. Additionally, the first phase of outlier treatment was applied directly to the transactional variables. Using the Interquartile Range (IQR) method, lower and upper thresholds were established as follows:

$$\text{Lower Bound} = Q_1 - 1.5 \times IQR \quad (1)$$

$$\text{Upper Bound} = Q_3 + 1.5 \times IQR \quad (2)$$

Outlier thresholds defined in Equations (1) and (2) were subsequently used for percentile-based winsorization to reduce distribution distortion while preserving the full customer population.

3.3.2 Feature Engineering

According to Zhou, et al., (2021), transactional data were aggregated by Customer ID to construct the RFMT model using four key variables: Recency (R), Frequency (F), Monetary (M), and Inter-purchase Time (T).

Recency (R): Number of days between the customer's most recent purchase and the dataset reference date ($D_{max} + 1$).

Frequency (F): Total number of unique invoices generated by the customer.

Monetary (M): Total revenue contributed by the customer:

$$M = \sum_{i=1}^n (Quantity_i \times Price_i) \quad (3)$$

Inter-purchase Time (T): The average time interval between customer purchases:

$$T = \frac{Date_{max} - Date_{min}}{F-1}, F > 1 \quad (4)$$

When $F = 1$, the value of T was assigned as the Recency value ($T = R$) to maintain data consistency. After constructing the RFMT features set, including the mathematical computation of Monetary and Inter-purchase Time defined in Equations (3) and (4), a final dataset of 5,878 unique customer profiles was constructed, with a sample of five representative records illustrated in Table 2. Because the aggregated RFMT distributions still exhibited significant variance, a secondary phase of outlier treatment was executed. The IQR-driven capping function was deployed across all four generated metrics to smooth the feature space.

Table 2. RFMT values of customers

Customer ID	Recency (days)	Frequency (times)	Monetary (£)	Inter-purchase Time (days)
12346	326	12	745.70	36.36
12347	2	8	4,921.53	57.43
12348	75	5	1,991.86	90.50
12349	19	4	3,786.50	190.00
12350	310	1	331.34	310.00

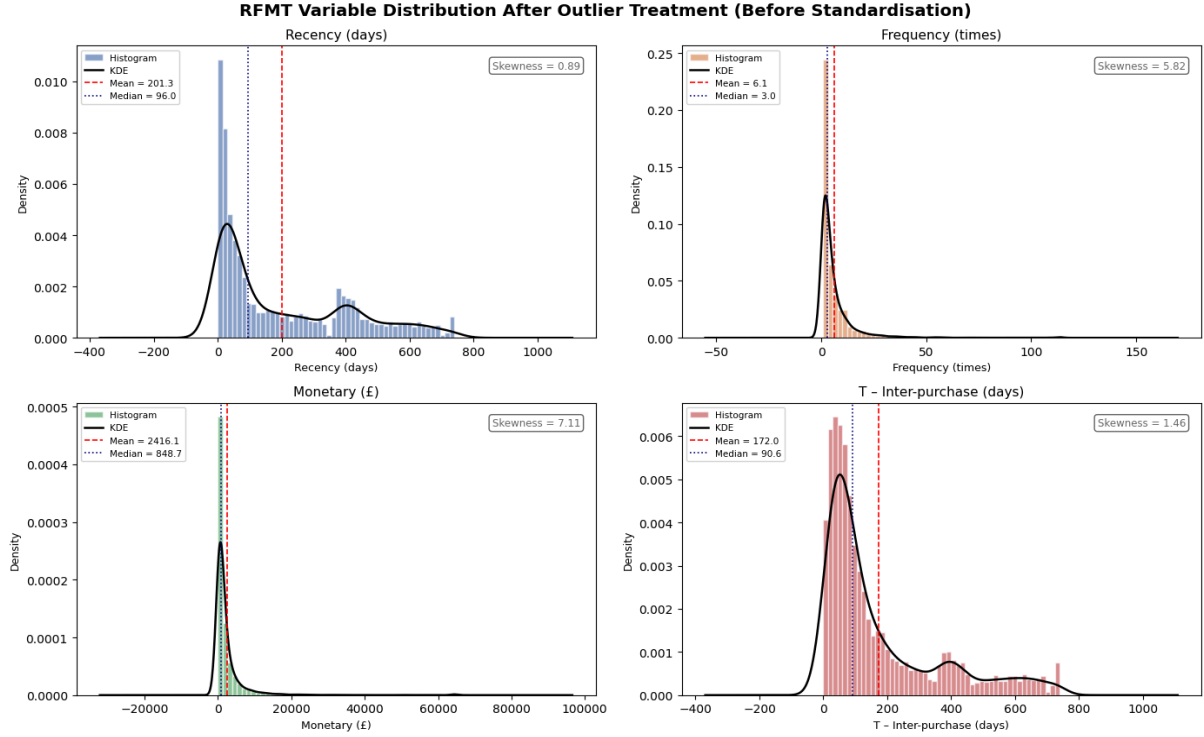


Fig. 2. Distribution of RFMT variables after outlier handling and before standardization

In Fig. 2, all variables exhibit right-skewed distributions. Due to the substantial differences in variable scales and the persistence of skewed distributions after outlier handling, the study further applies the StandardScaler method to standardize the data before performing data modeling.

3.4 Data Modeling

3.4.1 Target Variable Definition And Data Splitting

To construct the target variable, the dataset is divided into two temporal periods. The observation period spans from the start of the dataset up to 30 days before the final transaction date, which is utilized to compute the RFMT features. The final 30 days serve as the forecast period. The binary target Will_Return is labeled as 1 if a customer makes at least one transaction during the forecast period, and 0 otherwise. For evaluation, data is split into an 80% training set and a 20% testing set.

3.4.2 Baseline Classification Model

The baseline model performs prediction directly on the entire RFMT dataset without customer segmentation. According to Fig. 1, two experimental scenarios are implemented: one using the original imbalanced training data and another applying SMOTE-Tomek for class balancing. Four classification algorithms are deployed and compared: Logistic Regression, Random Forest, XGBoost, and Gradient Boosting.

3.4.3 Cluster-based Classification Model

In this pipeline, customer clustering is initially executed during the observation period using K-Means, Fuzzy C-Means, Spectral Clustering, and Mean Shift. Following segmentation,

separate classification models are developed tailored to each customer cluster. Consistent with the baseline track, the data within each cluster is split (80/20) and evaluated under both imbalanced and SMOTE-Tomek balanced scenarios using the same four classifiers.

3.5 Model Evaluation

Model performance was evaluated using multiple classification metrics, including Accuracy, Precision, Recall, F1-score, and ROC-AUC. While Accuracy was used to measure the overall prediction performance, it can be biased due to class imbalance. Therefore, Precision and Recall were employed to evaluate the model's effectiveness in identifying returning customers while minimizing false positive and false negative predictions. The F1-score, which represents the harmonic mean of Precision and Recall, was selected as the primary metric for final model selection. Additionally, the ROC-AUC is utilized to evaluate the models' overall discriminative capability across various probability thresholds.

4 Experimental Results And Discussion

4.1 Clustering Results

The optimal number of clusters for K-Means was determined using the Elbow Method and Silhouette Coefficient. As shown in Fig. 3(a), the distortion score displays a distinct inflection point at $k=5$. This choice is supported by Fig. 3(b), where the cluster configuration achieves an average silhouette score of 0.4523 with stable cluster assignments. This number of clusters was then applied to K-Means, Fuzzy C-Means (FCM), and Spectral Clustering for comparison, while Mean Shift was evaluated separately due to its density-based approach.

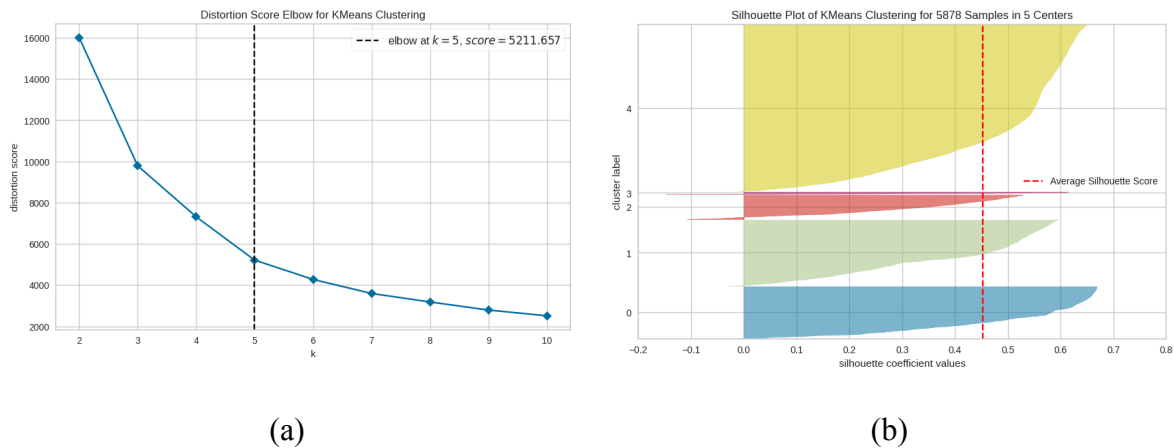


Fig. 3. Optimal cluster selection analysis: (a) Elbow Method; (b) Silhouette Score

Table 3. Cluster Size

Algorithm	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4
FCM	2835 (48%)	755 (13%)	47 (1%)	935 (16%)	1306 (22%)
K-Means	981 (17%)	1244 (21%)	464 (8%)	49 (1%)	3140 (53%)
Mean Shift	5809 (99%)	47 (1%)	22 (0%)	-	-
Spectral	3912 (67%)	84 (1%)	466 (8%)	359 (6%)	1057 (18%)

As shown in Table 4, Mean Shift achieved the highest Silhouette score (0.7504). However, the PCA visualization in Fig. 4 shows severe data concentration. According to Table 3, the algorithm grouped 99% of the customers into a single cluster with few outliers. This poor customer differentiation limits interpretability, making Mean Shift impractical for targeted retention strategies. Among the remaining methods, K-Means performed the best overall performance with the highest Silhouette score (0.4523) and a relatively balanced customer distribution across clusters. Conversely, Fuzzy C-Means showed similar results but produced less distinct segment boundaries because of its soft membership mechanism. Spectral Clustering recorded the lowest Silhouette score (0.3345), indicating weaker cluster separation.

Table 4. Silhouette Score for four clustering algorithms

Algorithm	k	Silhouette Score
K-Means	5	0.4523
Fuzzy C-Means	5	0.4325
Spectral Clustering	5	0.3345
Mean Shift	3	0.7504

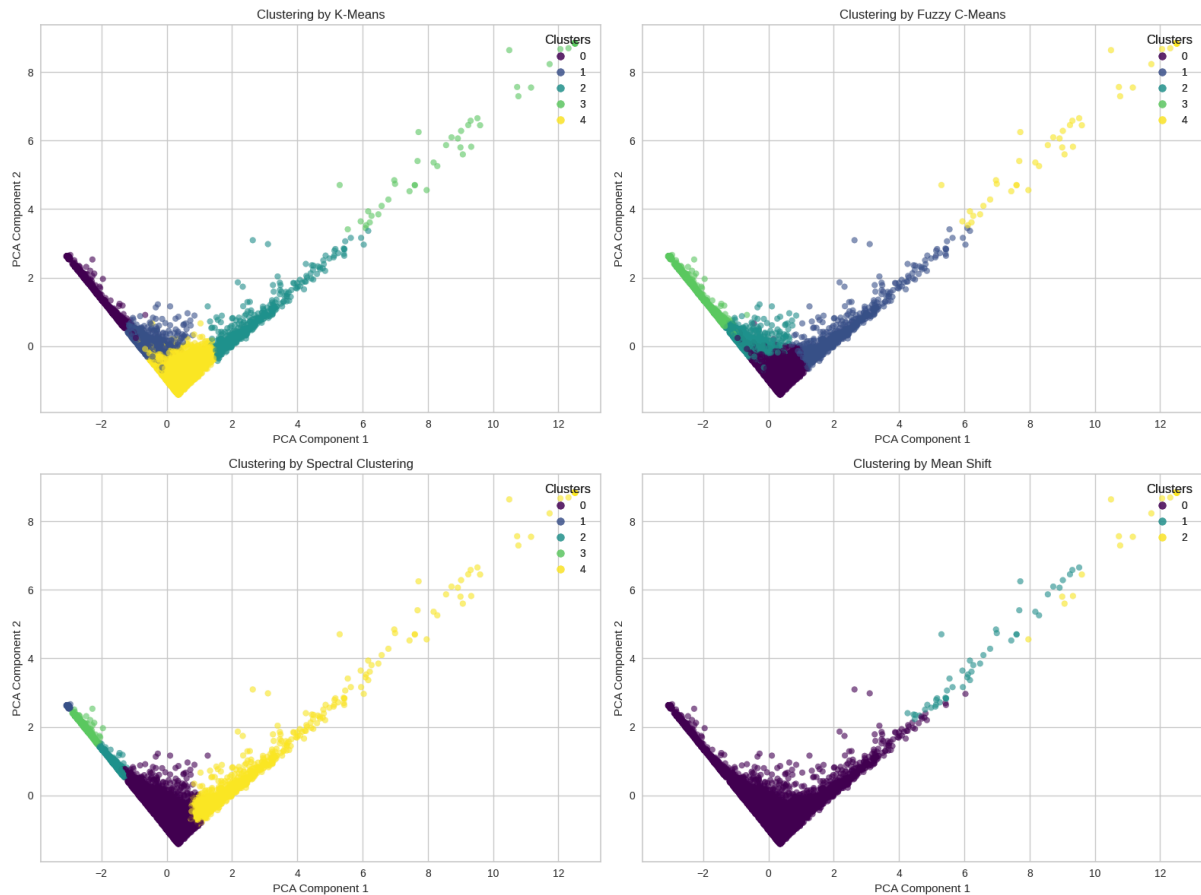


Fig. 4. PCA visualization of clustering results for the four evaluated algorithms

4.2 Classification Results

Table 5 displays classification performance between the global baseline approach and the cluster-based pipeline under both imbalanced and SMOTE-Tomek-balanced settings. Without class balancing, all global classifiers achieved relatively high Accuracy (0.74-0.78) but low Recall (0.30-0.39), indicating strong bias toward the majority class. After applying SMOTE-Tomek, Recall improved substantially across all models, although Accuracy decreased slightly. Logistic Regression achieved the best global F1-score (0.5310), followed by XGBoost (0.5117).

Table 5. Consolidated classification results: global baseline vs. cluster-based (with and without SMOTE-Tomek)

Classifier	Accuracy	Precision	Recall	F1-score	ROC-AUC
Global without SMOTE-Tomek					
Logistic Regression	0.7815	0.6875	0.2953	0.4131	0.7620
Random Forest	0.7430	0.5088	0.3893	0.4411	0.7242
Gradient Boosting	0.7771	0.6443	0.3221	0.4295	0.7505
XGBoost	0.7754	0.6242	0.3456	0.4449	0.7444
Global with SMOTE-Tomek					
Logistic Regression	0.6696	0.4213	0.7181	0.5310	0.7621
Random Forest	0.6783	0.4175	0.5940	0.4903	0.7231
Gradient Boosting	0.6678	0.4142	0.6644	0.5103	0.7427
XGBoost	0.6713	0.4174	0.6611	0.5117	0.7376
Cluster-Based without SMOTE-Tomek					
K-Means + Logistic Regression	0.8572	0.6375	0.5400	0.5396	0.6531
K-Means + Random Forest	0.8153	0.5570	0.5648	0.5591	0.6773
K-Means + Gradient Boosting	0.8499	0.6292	0.5354	0.5548	0.7800
K-Means + XGBoost	0.8182	0.5842	0.5372	0.5448	0.7487
FCM + Logistic Regression	0.8319	0.6423	0.5198	0.5207	0.6114
FCM + Random Forest	0.8059	0.5471	0.5271	0.5293	0.7321
FCM + Gradient Boosting	0.8131	0.6234	0.4845	0.5153	0.7059
FCM + XGBoost	0.8098	0.5721	0.5164	0.5182	0.6913
Spectral + Logistic Regression	0.7303	0.3341	0.5000	0.4006	0.5686
Spectral + Random Forest	0.7142	0.5587	0.5065	0.5173	0.6816
Spectral + Gradient Boosting	0.7312	0.6717	0.4424	0.4539	0.7162
Spectral + XGBoost	0.7300	0.6023	0.4471	0.4789	0.6786
Cluster-Based with SMOTE-Tomek					
K-Means + Logistic Regression	0.7088	0.5853	0.7911	0.5710	0.7763
K-Means + Random Forest	0.7229	0.5561	0.6283	0.5454	0.7527
K-Means + Gradient Boosting	0.7286	0.5836	0.7639	0.5647	0.8044
K-Means + XGBoost	0.7692	0.5966	0.7893	0.5901	0.8247

FCM + Logistic Regression	0.6950	0.5657	0.7246	0.5261	0.7616
FCM + Random Forest	0.7663	0.5253	0.5140	0.5176	0.6755
FCM + Gradient Boosting	0.7645	0.5508	0.5136	0.5304	0.6580
FCM + XGBoost	0.7927	0.5534	0.5441	0.5481	0.7024
Spectral + Logistic Regression	0.5926	0.4717	0.6831	0.5439	0.6025
Spectral + Random Forest	0.6669	0.5437	0.5580	0.5438	0.6920
Spectral + Gradient Boosting	0.6507	0.5542	0.6247	0.5636	0.7052
Spectral + XGBoost	0.6661	0.5450	0.6441	0.5764	0.6950

* *Best-performing model*

For the cluster-based approach, a separate classifier was trained within each customer segment. Among clustering methods, K-Means consistently produced the best results, outperforming Fuzzy C-Means and Spectral Clustering. Spectral Clustering showed the weakest performance due to poor class distribution in several segments. After applying SMOTE-Tomek within each cluster, performance improved across all segmentation methods. The hybrid K-Means-XGBoost model with SMOTE-Tomek achieved the best overall performance, with an F1-Score of 0.5901 and ROC-AUC of 0.8247. Compared to the best-performing global baseline (Logistic Regression with SMOTE-Tomek), the model improved F1-score by 11.13%. Notably, the ROC-AUC improved from 0.7621 to 0.8247, suggesting that the cluster-based framework achieved substantially better class separability and overall discriminative performance. These findings suggest that customer segmentation helps reduce behavioral differences within each group, allowing classification models to identify clearer decision patterns for each segment.

4.3 Best-Performing Model Analysis And Business Insights

The hybrid K-Means-XGBoost model integrated with the SMOTE-Tomek resampling technique demonstrates strong capability in predicting customer return behavior, particularly for identifying customers unlikely to return. Based on the confusion matrix in Fig. 5, the model correctly identified 436 out of 599 non-returning customers, achieving a recall rate of 72.79% for the Not Return class, indicating high effectiveness in detecting customers at risk of disengagement. In addition, the model correctly predicted 178 out of 297 returning customers, corresponding to a recall rate of 59.93% for the Will Return class. These results suggest that the model maintains relatively balanced predictive performance across both customer groups while showing stronger sensitivity toward identifying non-returning customers. From a business perspective, the ability to recognize customer repurchase propensity allows firms to identify early signs of disengagement. This predictive insight enables organizations to deploy timely retention strategies, targeted marketing campaigns, and proactive incentives that reinforce repeat buying behavior, thereby mitigating churn risks and improving long-term loyalty.

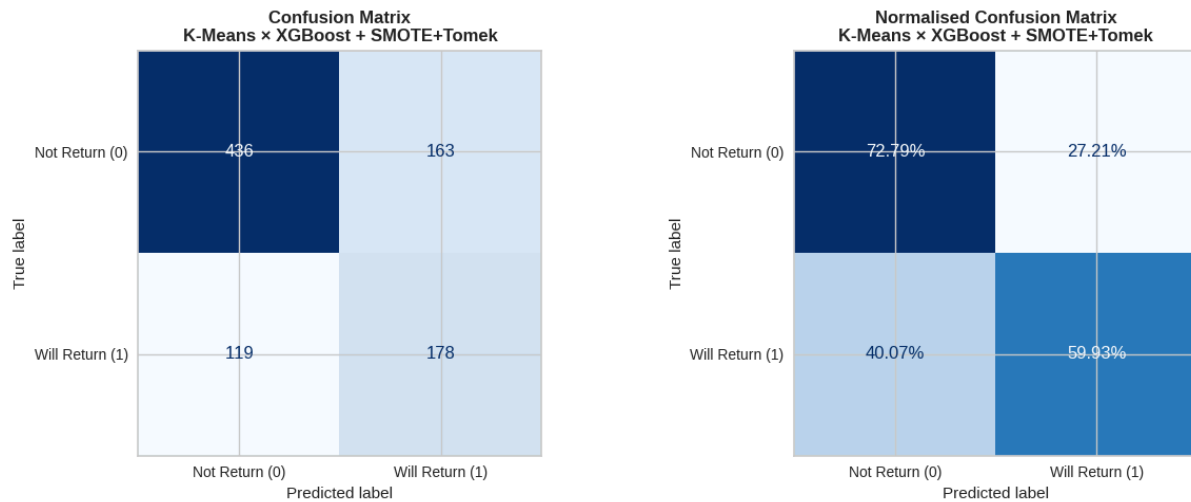


Fig. 5. Confusion matrix of the hybrid K-Means -XGBoost model with SMOTE-Tomek

The charts in Fig. 6 reveal clear behavioral differences across customer groups based on RFMT characteristics. VIPs exhibit the strongest purchasing behavior, with near-weekly purchasing ($T=10.5$ days) and wholesale-level spend, signalling habitual, operationally-driven buying. Potential Loyalists also show strong purchasing and retention behavior. Their 33-day repurchase interval mirrors a monthly buying rhythm that, if reinforced, could shorten toward the VIPs cadence. This segment offers the highest loyalty upgrade potential; without proactive engagement, natural churn will migrate a portion toward Occasional Buyers. With Occasional Buyers, they tend to purchase roughly quarterly ($T=102.9$ days). The gap between their Monetary value (£1,683) and Potential Loyalists (£10,573) suggests limited product exploration. At Risk customers made a single, small purchase approximately 17 months ago and never returned. The $T=527$ -day figure is an artefact of single-purchase behaviour rather than a true repurchase interval, indicating that no buying relationship was ever established. Unlike At Risk customers, the Churned ones had a genuine purchasing relationship historically with multiple quarterly purchases and nearly £1,000 in historical spend. Their primary analytical value is diagnostic: Occasional Buyers with declining Frequency and rising Recency indicate declining engagement and warrant pre-emptive intervention.

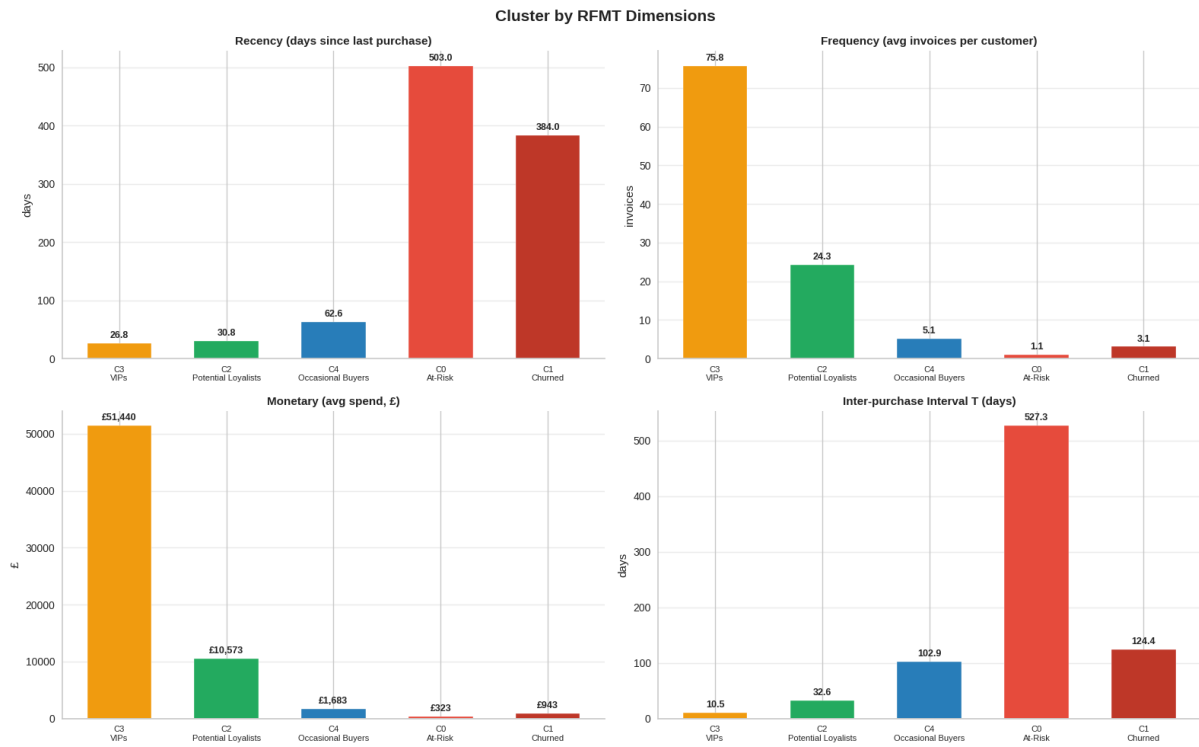


Fig. 6. Cluster profiles by RFMT Dimensions

Figs. 7 and 8 show a notable insight into the proportion of cluster size and revenue contribution and their return rate. Although VIPs and Potential Loyalists represent less than 10% of total customers, they contribute nearly 60% of total revenue. This highlights the strategic importance of retaining and developing high-value customer segments. The Occasional Buyers, while comprising over 50% of the customer base, reveals only a moderate return rate of 36.5%, suggesting that further breakdown is needed to leverage the return rate of the promising segment. Conversely, At Risk and Churned account for nearly 38% of total customers, but only contribute approximately 9.2% of the remaining revenue. This indicates that these two groups have low economic value and limited exploitation potential. Businesses need to consider the trade-off between reactivation costs and expected benefits when implementing retention campaigns.

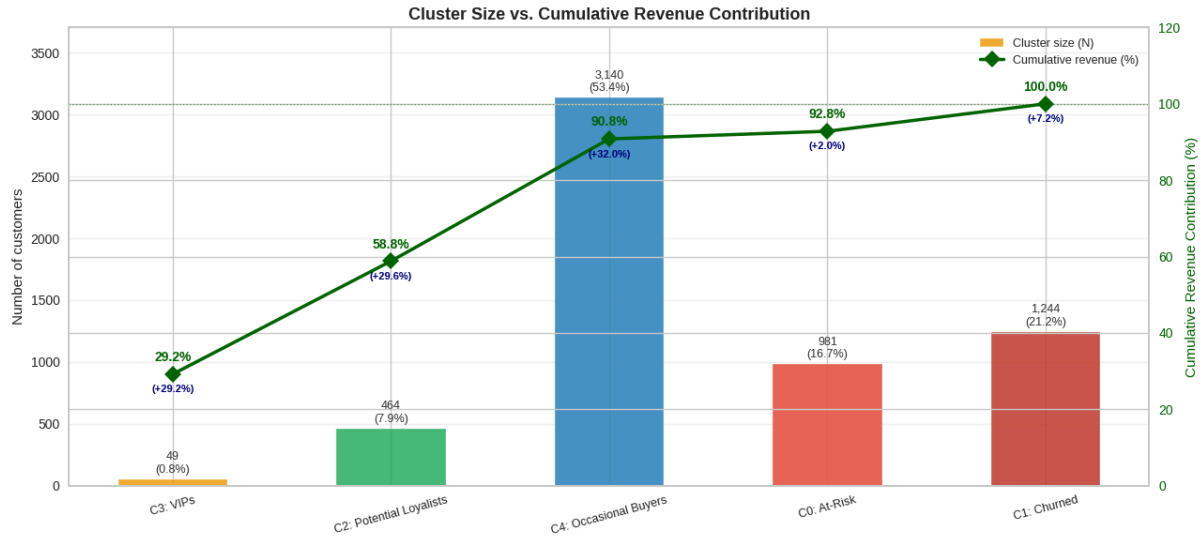


Fig. 7. Cluster Size vs. Cumulative Revenue Contribution

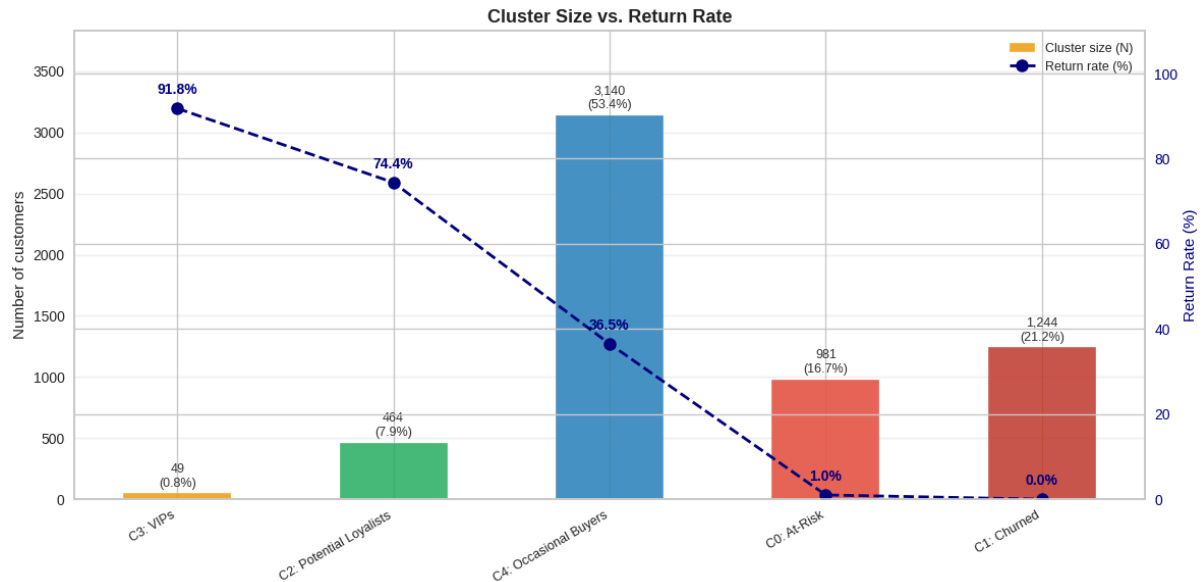


Fig. 8. Cluster Size vs. Return Rate

Overall, the integration of customer segmentation with predictive modeling provides both strong predictive performance and meaningful business interpretability. The results demonstrate that combining behavioral clustering with repurchase prediction enables organizations to identify economically important customer groups while supporting more targeted and proactive retention strategies.

5 Conclusion

The experimental results showed that cluster-based prediction generally outperformed the global baseline approach, particularly after applying SMOTE-Tomek balancing. Among all evaluated models, the hybrid K-Means-XGBoost model with SMOTE-Tomek achieved the best overall performance, with an F1-score of 0.5901 and ROC-AUC of 0.8247, consistent with the inherent difficulty of repurchase prediction in the retail e-commerce sector. The

findings also demonstrated that high-value customer groups contributed disproportionately to total revenue despite representing a relatively small portion of the customer base, emphasizing the need for segment-specific retention strategies over broad reactivation campaigns. Despite these promising results, the study still has several limitations. First, the experiments were conducted without cross-validation, which may limit the robustness and generalizability of the reported classification performance. Future work should incorporate k-fold cross-validation or stratified cross-validation to provide more reliable model evaluation. Second, for consistency and comparison purposes, a fixed cluster size was enforced for all algorithms, with the exception of Mean Shift. However, the optimal number of clusters may differ depending on the characteristics and assumptions of each algorithm. Future research could explore adaptive or algorithm-specific cluster selection methods to further improve segmentation quality and predictive performance. Third, the 30-day forecast window serves as a behavioral proxy for repurchase likelihood and may not fully capture long-term disengagement patterns, particularly for segments with longer inter-purchase cycles such as Occasional Buyers ($T=102.9$ days). Future research could explore adaptive or dynamic forecast windows calibrated to each segment's purchasing rhythm. Overall, this research demonstrates the potential of combining customer segmentation and predictive analytics to support data-driven customer relationship management and personalized marketing strategies in retail businesses.

References

1. Ahmed, H. H., Khafagy, M. H., & Kaseb, M. R. (2024). A Novel Model for Partial and Total Churn Prediction in E-Commerce. *Journal of Electrical Systems*, 20(3), 180–195.
2. Almuqren, L., Alrayes, F. S., & Cristea, A. I. (2021). An Empirical Study on Customer Churn Behaviours Prediction Using Arabic Twitter Mining Approach. *Future Internet*, 13(7), 175.
3. Bose, I., & Chen, X. (2009). Hybrid Models Using Unsupervised Clustering for Prediction of Customer Churn. *Journal of Organizational Computing and Electronic Commerce*, 19(2), 133-151.
4. Chakraborty, S., Paul, D., & Das, S. (2021, May). Automated clustering of high-dimensional data with a feature weighted mean shift algorithm. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 35, No. 8, pp. 6930-6938).
5. Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321–357.
6. Chen, D. (2015). *Online Retail II* [Data set]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5BW33>
7. Ding, Z., Wang, Z., Hu, T., & Wang, H. (2022). A Comprehensive Study on Integrating Clustering with Regression for Short-Term Forecasting of Building Energy Consumption: Case Study of a Green Building. *Buildings*, 12(10), 1701.
8. Dua, S., Kumar, D., Dua, S., & Batra, J. (2025, July). A Hybrid Approach to Optimize Market Segmentation: Integration of Unsupervised Clustering and Supervised Ensemble Method. In *Recent Advances in Artificial Intelligence for Sustainable Development (RAISD 2025)* (pp. 474-483). Atlantis Press.

9. Ekanayake, I., & De Alwis, S. (2025). Explainability, risk modeling, and segmentation based customer churn analytics for personalized retention in e-commerce. *arXiv preprint arXiv:2510.11604*.
10. Gan, L. (2022). XGBoost-Based E-Commerce Customer Loss Prediction. *Computational Intelligence and Neuroscience*, 2022(1), 1858300.
11. Gankidi, N., Gundu, S., viqar Ahmed, M., Tanzeela, T., Prasad, C. R., & Yalabaka, S. (2022, June). Customer segmentation using machine learning. In *2022 2nd International Conference on Intelligent Technologies (CONIT)* (pp. 1-5). IEEE.
12. Hughes, A. M. (2005). *Strategic database marketing*. McGraw-Hill Pub. Co.
13. Jahromi, A. T., Stakhovych, S., & Ewing, M. (2014). Managing B2B customer churn, retention and profitability. *Industrial Marketing Management*, 43(7), 1258–1268.
14. Li, X., & Li, Z. (2019). A Hybrid Prediction Model for E-Commerce Customer Churn Based on Logistic Regression and Extreme Gradient Boosting Algorithm. *Ingénierie des Systèmes d'Information*, 24(5).
15. Liu, R., Ali, S., Bilal, S. F., Sakhawat, Z., Imran, A., Almuhaimeed, A., ... & Sun, G. (2022). An Intelligent Hybrid Scheme for Customer Churn Prediction Integrating Clustering and Classification Algorithms. *Applied Sciences*, 12(18), 9355.
16. Martínez, A., Schmuck, C., Pereverzyev Jr, S., Pirker, C., & Haltmeier, M. (2020). A machine learning framework for customer purchase prediction in the non-contractual setting. *European Journal of Operational Research*, 281(3), 588-596.
17. Nguyen, T. D., & Bui, T. H. (2024, November). Predicting Customer Attrition in Claims and Direct Billing Services of Health Insurance. In *The 2nd International Conference: Resiliency by Technology and Design (RTD 2024)* (pp. 450-471). Atlantis Press.
18. Praseeda, C. K., & Shivakumar, B. L. (2021). Fuzzy particle swarm optimization (FPSO) based feature selection and hybrid kernel distance based possibilistic fuzzy local information C-means (HKD-PFLICM) clustering for churn prediction in telecom industry. *SN Applied Sciences*, 3(6), 613.
19. Rahul, S., Salokhe, L., & Jadhav, S. (2021). Customer segmentation using RFM model and k-means clustering. *International Journal of Scientific Research in Science and Technology*, 8(3), 591-597.
20. Tang, Q., Xia, G., & Zhang, X. (2020). A hybrid classification model for churn prediction based on customer clustering. *Journal of Intelligent & Fuzzy Systems*, 39(1), 69-80.
21. Tomek, I. (1976). Two modifications of CNN. *IEEE Transactions on Systems, Man, and Cybernetics*, SMC-6(11), 769-772.
22. Ullah, A., Mohmand, M. I., Hussain, H., Johar, S., Khan, I., Ahmad, S., Mahmoud, H. A., & Huda, S. (2023). Customer Analysis Using Machine Learning-Based Classification Algorithms for Effective Segmentation Using Recency, Frequency, Monetary, and Time. *Sensors*, 23(6), 3180.
23. Verma, R., Rathor, D., Kumar, S., Mishra, M., & Baranwal, M. (2025). Enhancing customer repurchase prediction: integrating classification algorithms with RFM analysis for precision and actionable insights. *IIMB Management Review*, 37(2), 100574.

24. Wu, Z., Jin, L., Zhao, J., Jing, L., & Chen, L. (2022). Research on Segmenting E-Commerce Customers through an Improved K-Medoids Clustering Algorithm. *Computational Intelligence and Neuroscience*, 2022, 9930613.
25. Xiahou, X., & Harada, Y. (2022). B2C E-Commerce Customer Churn Prediction Based on K-Means and SVM. *Journal of Theoretical and Applied Electronic Commerce Research*, 17(2), 458-475.
26. Zhou, J., Wei, J., & Xu, B. (2021). Customer segmentation by web content mining. *Journal of Retailing and Consumer Services*, 61, 102588.

Data And Code Availability

Code and supplementary materials are publicly available at:

<https://github.com/eve-nguyn/customer-segmentation-repurchase-prediction/>